I. Project Planning and Task Definition

During the project planning phase (Fall Semester, 2025) many of the project members had conflicting schedules and could not regularly meet. A significant portion of the work done during this phase was research and conceptual design, but hardware design, testing, and software design would not commence until Spring 2026. Due to this significant delay the project was scheduled and executed under severe time constraints.

Once regular contact was established between members, meetings were held each Monday from 11:00-2:00, 4:00-5:00 PM, and often on Wednesday at the same time from 11:00-2:00. Brandon chose to focus on the software that would be implemented for the SCADA system, testing it on both the Raspberry Pi and a laptop using several Arduino boards. This was done rather sporadically as bugs were regularly encountered and resolved. This effort was eventually unified with the rest of the project when C++ code generation was implemented. Because this task was relatively insular he did not interact with the team as regularly as the other members.

Karina, Steven and Carlos focused on leading the hardware effort, testing and adjusting circuitry and sensor readings using the test code Karina developed in parallel to the SCADA code (pre code-gen). The test code developed during these experiments informed how the code generator Brandon developed would have to work. Karina managed a schedule using Gantt charts, one of which is posted below. Marlene helped secure the fittings onto the container resulting in water-tight connections, as well as working on the spigot (for manual emptying) and the valves.

	Wednesday 3/18/2026	Sunday 3/22/2026	Monday 3/23/2026
Control Circuit	Start Building, issues with 12v supply		
Pump Control Code	Test it, works		
Water level sensor code (A)		Got min and max values, mounted	
Water level sensor code (D)			
LED indicators			
Power Bus	Switch test		
Flow Rate Sensor Code			
Flow Rate sensor Installation	Done		
Pump Hardware Test	Done		
Valve Hardware Test	Done		
Spigot	Research	Drilled hole, leaky	
Tank Water management	Research		
Water proof sealing	Ordered Silicon	Got it waiting	
Water level sensor test (D)			
Documentation			

Fig. 1. Gantt Chart used for Event Scheduling

Steven taught Carlos how to solder, as well as building the initial schematic that governed the arrangement of MOSFTs and other circuit components, eventually soldering them onto prototype board. It was eventually deemed that commercially available power circuits would be used for convenience. Marlene put a substantial effort into graphic design, producing the Symposium Poster used by our group.